



United International University

Department of Computer Science and Engineering

Summer 2024 CSE 3411 System Analysis and Design

Assignment - 01

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EzyShop Online is a robust e-commerce platform designed to provide users with a seamless and personalized shopping experience. Catering to individual consumers, wholesale buyers, and vendors, EzyShop offers a range of features such as product browsing, personalized recommendations, secure checkout, and multiple payment options. Users can create accounts or log in to access enhanced features, with authentication managed through the USER database. To enhance security, if a user fails authentication three times, their account is temporarily locked.

Once logged in, users can search for products using keywords, filters, or browse through various categories, with the system fetching relevant details from the PRODUCT_CATALOG database. The platform uses advanced machine learning algorithms to offer personalized product recommendations based on users' past purchases, browsing history, and items saved in their wishlists. Users can add items to their shopping cart, modify quantities, or remove products as needed, with the cart information being saved in the SHOPPING_CART database.

During the checkout process, the system verifies product availability from the INVENTORY database and calculates the total cost, including applicable taxes and shipping fees. Users can choose from various payment methods, including credit/debit cards, digital wallets, and Buy Now, Pay Later options. Upon successful payment, the ORDER_MANAGEMENT system generates an order confirmation and updates the INVENTORY and ORDER databases accordingly. Users receive an order confirmation email with details of their purchase and estimated delivery timelines. If payment is not completed within a specified timeframe, the order is automatically canceled, and the items are returned to inventory.

For vendors, EzyShop provides a seller portal where they can list products, manage stock, and track sales performance. Vendors can also view detailed analytics on customer interactions and feedback, helping them optimize their offerings. The platform aims to create a comprehensive, user-friendly shopping environment that caters to the needs of both buyers and sellers, enhancing engagement and driving growth in the online marketplace.

Questions:

- a. Draw the Data Flow Diagram (DFD) for the above scenario. [3]
- b. Draw the Use Case Diagram for the above scenario (include at least one extend/include [4] relationship in the diagram). Write the descriptive form of one major use case shown in the use case diagram for the above scenario.
- c. Draw the Activity & Swimlane Diagram. [3]